

飞雄光电

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SPECIFICATION FOR LCD MODULE

MODULE NO.:F55H40-152
DOC.REVISION :V0

Customer Approval:

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	SIGNATURE	DATE
PREPARED BY		2023.03.20
CHECKED BY		
APPROVED BY		

DOCUMENT REVISION HISTORY

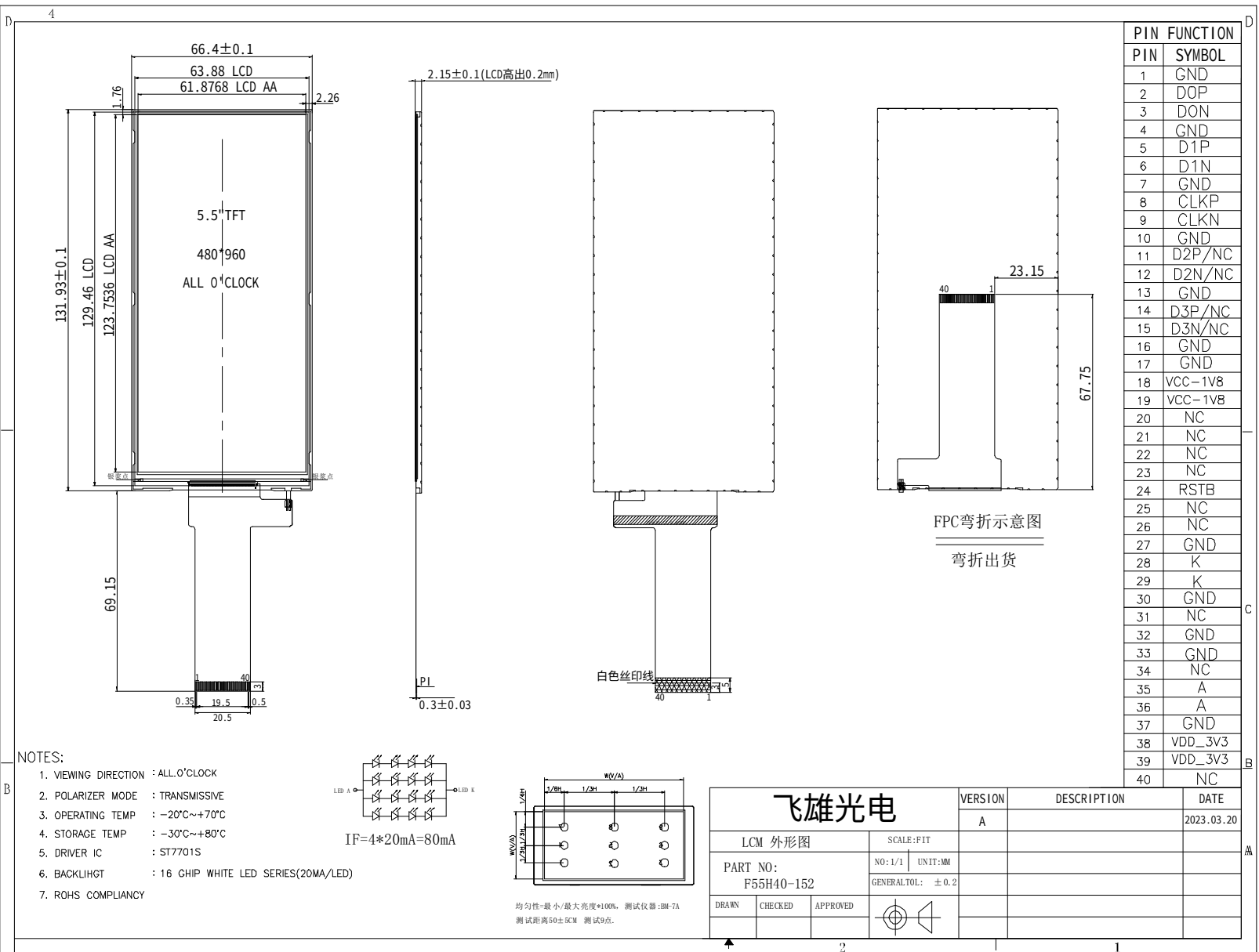
Version	DATE	DESCRIPTION	CHANGED BY
V0	2023.03.20	New design	

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1. Features & Mechanical Specifications

Item	Contents	Unit
	LCD	
LCD Type	TFT Transmissive Normal Black	--
Viewing direction	ALL.O'CLOCK	--
Backlight	White LED x16in Parallel/ Series	--
Interface	MIPI 2LANES	--
Driver IC	ST7701S	--
Outline Dimension	66.40*131.93*2.15	mm
Glass area (W×H×T)	63.884 (H) x 129.458 (V) x 0.5 (T) (Typ.)	mm
Active area (W×H)	61.8768 (H) x 123.7536 (V)	mm
Number of Dots	480 RGB (H) x 960 (V)	pixels
Pixel pitch (W×H)	0.12891 (H) x 0.12891 (V)	mm
Operating Temperature	-20 ~ +70	℃
Storage temperature	-30 ~ +80	℃

2. Dimensional Outline



3. Pin Description

PIN No.	SYMBOL	Function
1	GND	Ground
2	DSI-D0P	DSI Data differential signal input pins. (Data lane0)
3	DSI-D0N	DSI Data differential signal input pins. (Data lane 0)
4	GND	Ground
5	DSI-D1P	DSI Data differential signal input pins. (Data lane 1)
6	DSI-D1N	DSI Data differential signal input pins. (Data lane 1)
7	GND	Ground
8	DSI-CLK P	DSI CLOCK differential signal input pins
9	DSI-CLK N	DSI CLOCK differential signal input pins
10	GND	Ground
11-12	NC	NC
13	GND	Ground
14-15	NC	NC
16-17	GND	Ground
18-19	IOVCC	I/O Power supply
20-23	NC	NC
24	RESET	Reset Signal pin (“Low” is enable)
25-26	NC	NC
27	GND	Ground
28-29	LEDK	Backlight LED Cathode
30	GND	Ground
31	NC	NC
32-33	GND	Ground
34	NC	NC
35-36	LEDA	Backlight LED Anode.
37	GND	Ground
38-39	VCC	Logic Power supply
40	NC	NC

4. Absolute Maximum Ratings

Item	Symbol	Rating			Unit
		MIN.	TYP.	MAX	
Operating Temperature range	TOP	-20	-	+70	°C
Storage Temperature range	TST	-30	-	+80	°C

5. Electrical Characteristics

DC Characteristics

Item	Symbol	Min.	Type.	Max.	Unit
Logic Supply Voltage	VCC	2.8	-	3.3	V
I/O Supply Voltage	IOVCC	1.8	-	3.3	V

注意：复位 RESET 电压必须与 IOVCC 保持一致，没有 IOVCC 时复位电压必须与 VCC 保持一致。

6. Backlight Characteristics

White LED × 16 in Parallel/ Series

(Ta = 25°C)

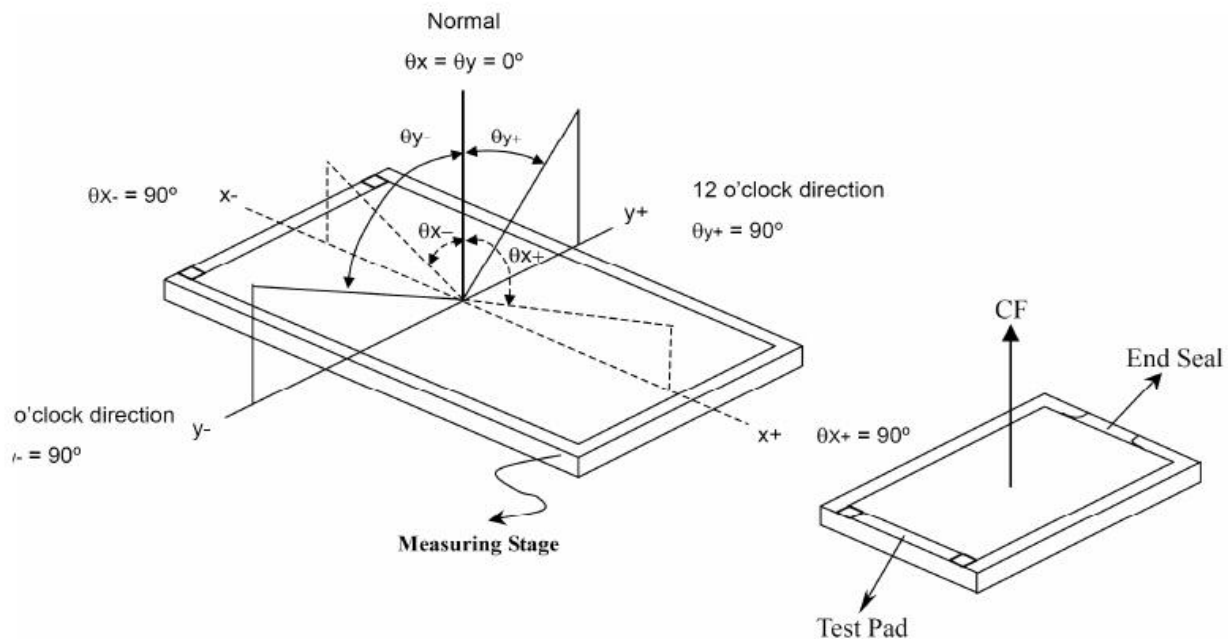
Item	Symbol	Condition	Min	Typ	Max	Unit
Forward Voltage	VF	IF=80mA	11.2	-	13.2	V
Uniformity	△Bp	-	80	-	-	%
Luminance for LCD	Lv	IF=80mA	650	700	-	cd/m ²

7. Electro-Optical Characteristics

Using LC+ Normal Polarizer+Corresponding Backlight, reference only (Note 1,Note 2)

Parameter		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
Viewing Angle range	Horizontal	Θ_3	CR > 10	-	80	-	Deg.	Note 1
		Θ_9		-	80	-	Deg.	
	Vertical	Θ_{12}		-	80	-	Deg.	
		Θ_6		-	80	-	Deg.	
Contrast ratio		CR	$\Theta = 0^\circ$	-	800	-		Note 2
Transmittance		Tr		-	4.5	-	%	Base on C Light Note 3
White Chromaticity		x_w		-	0.298	-		Note 4 CF Glass Base on C Light
		y_w		-	0.328	-		
Reproduction of color (C light)	Red	R_x		-	0.659	-		
		R_y		-	0.322	-		
	Green	G_x		-	0.290	-		
		G_y		-	0.588	-		
	Blue	B_x		-	0.134	-		
		B_y		-	0.124	-		
Response Time (Rising + Falling)		$T_r + T_f$	Ta= 25° C $\Theta = 0^\circ$	-	30	35	ms	Note 5

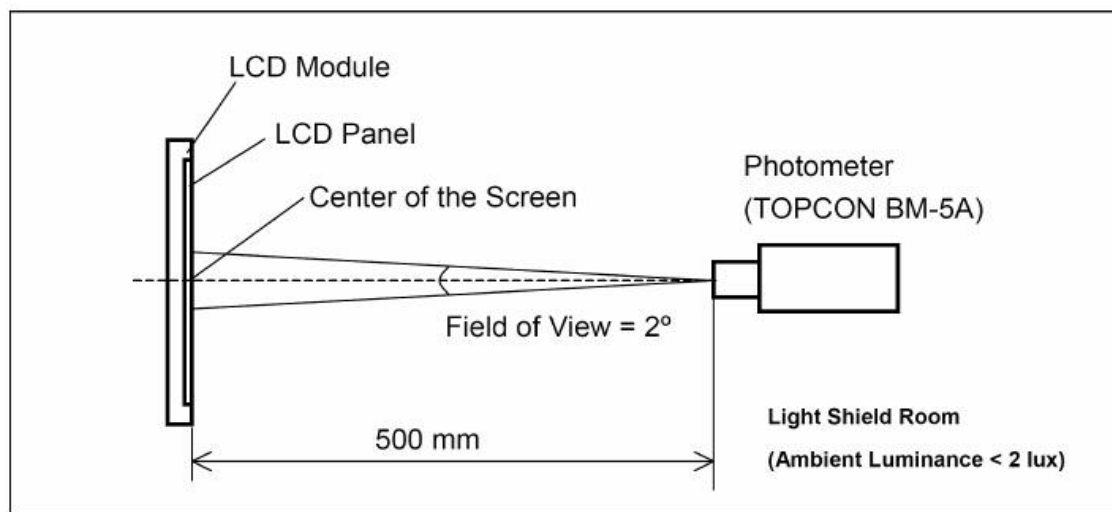
***Note(3) Definition of Viewing Angle**



*** The above "Viewing Angle" is the measuring position with Largest Contrast Ratio; not for good image quality. View Direction for good image quality is 6 O'clock. Module maker can increase the "Viewing Angle" by applying Wide View Film.

***Note (4) Measurement Set-Up:**

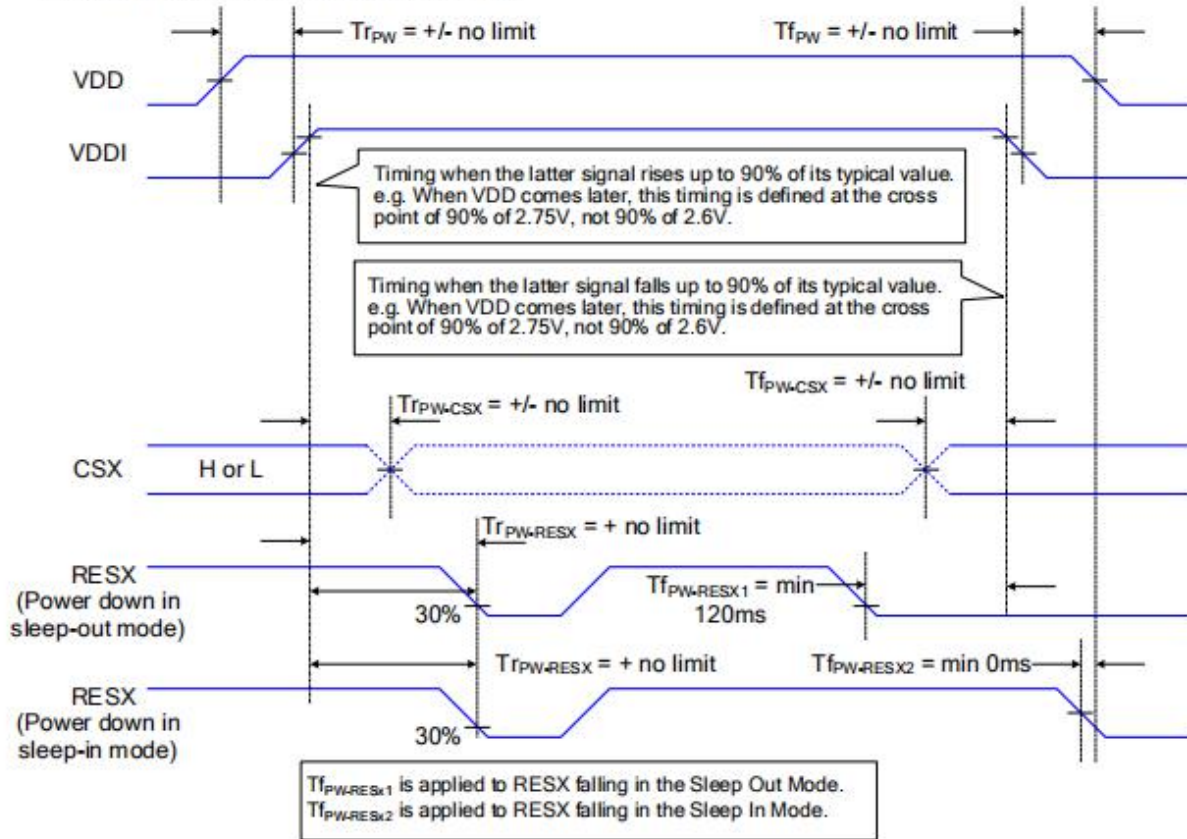
The LCD module should be stabilized at a given temperature for 20 minutes to avoid abrupt temperature change during measuring. In order to stabilize the luminance, the measurement should be executed after lighting Backlight for 20 minutes in a windless room.



8. Instruction Description

Please refer to **ST7701S**

The power on/off sequence is illustrated below



VDDI and VDDA can be applied or powered down in any order. During the Power Off sequence, if the LCD is in the Sleep Out mode, VDDA and VDDI must be powered down with minimum 120msec. If the LCD is in the Sleep In mode, VDDA and VDDI can be powered down with minimum 0msec after the RESX is released.

CSX can be applied at any timing or can be permanently grounded. RESX has high priority over CSX.

Notes:

1. There will be no damage to the ST7701S if the power sequences are not met.
2. There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.
3. There will be no abnormal visible effects on the display between the end of Power On Sequence and before receiving the Sleep Out command, and also between receiving the Sleep In command and the Power Off Sequence.
4. If the RESX line is not steadily held by the host during the Power On Sequence as defined in Sections 8 , then it will be necessary to apply the Hardware Reset (RESX) after the completion of the Host Power On Sequence to ensure correct operations. Otherwise, all the functions are not guaranteed.

The power on/off sequence is illustrated below

The uncontrolled power-off means a situation which removed a battery without the controlled power off sequence. It will neither damage the module or the host interface.

If uncontrolled power-off happened, the display will go blank and there will not any visible effect on the display (blank display) and remains blank until "Power On Sequence" powers it up.

9. AC Characteristics

MIPI DSI Timing Characteristics HIGH SPEED MODE

MIPI Interface Characteristics:

High Speed Mode

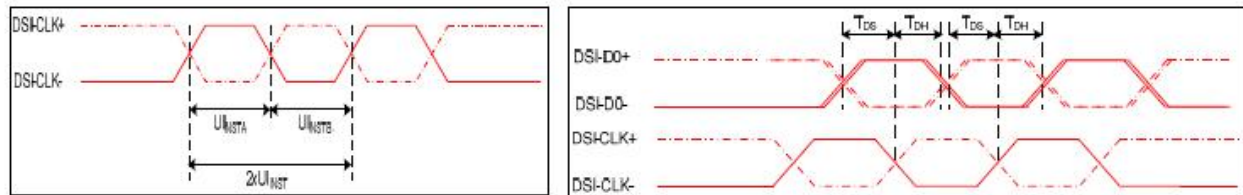


Figure 4 DSI clock channel timing

Figure 5 Rising and falling time on clock and data channel

$VDDI=1.8, VDD=2.8, AGND=DGND=0V, T_a=25^{\circ}C$

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
DSI-CLK+/-	$2 \times UI_{INSTA}$	Double UI instantaneous	4	25	ns	
DSI-CLK+/-	UI_{INSTA} UI_{INSTB}	UI instantaneous halves	2	12.5	ns	$UI = UI_{INSTA} = UI_{INSTB}$
DSI-Dn+/-	tDS	Data to clock setup time	0.15	-	UI	
DSI-Dn+/-	tDH	Data to clock hold time	0.15	-	UI	

Table 7 Mipi Interface- High Speed Mode Timing Characteristics

10.Quality Specifications

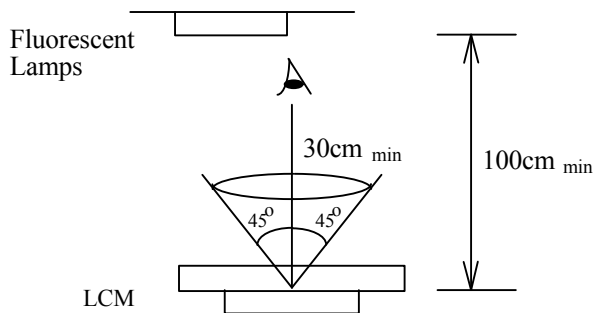
All The raw material are Rohs complicant.

10.1 Standard of the product appearance test

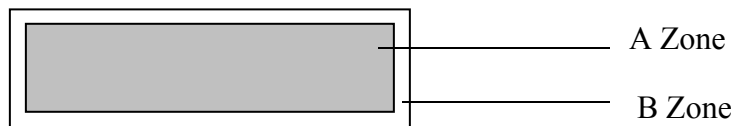
Manner of appearance test: The inspection should be performed in using 20W x 2 fluorescent lamps.

Distance between LCM and fluorescent lamps should be 100 cm or more. Distance between LCM and inspector eyes should be 30 cm or more.

Viewing direction for inspection is 45° from vertical against LCM.



Definition of zone:



A Zone: viewing area

B Zone: outside viewing area

10.2 Reliability of LCM

Reliability test condition:

Item	Condition	Time (hrs)	Assessment
High temp. Storage	80°C	48	No abnormalities in functions and appearance
High temp. Operating	70°C	48	
Low temp. Storage	-30°C	48	
Low temp. Operating	-20°C	48	
Humidity	60°C/ 90%RH	48	
Temp. Cycle	-30°C ← 25°C → 80°C (60 min ← 5 min → 60min)	10cycles	

Recovery time should be 24 hours minimum. Moreover, functions, performance and appearance shall be free from remarkable deterioration within 50,000 hours under ordinary operating and storage conditions room temperature ($20 \pm 8^{\circ}\text{C}$), normal humidity (below 65% RH), and in the area not exposed to direct sun light.

- END -